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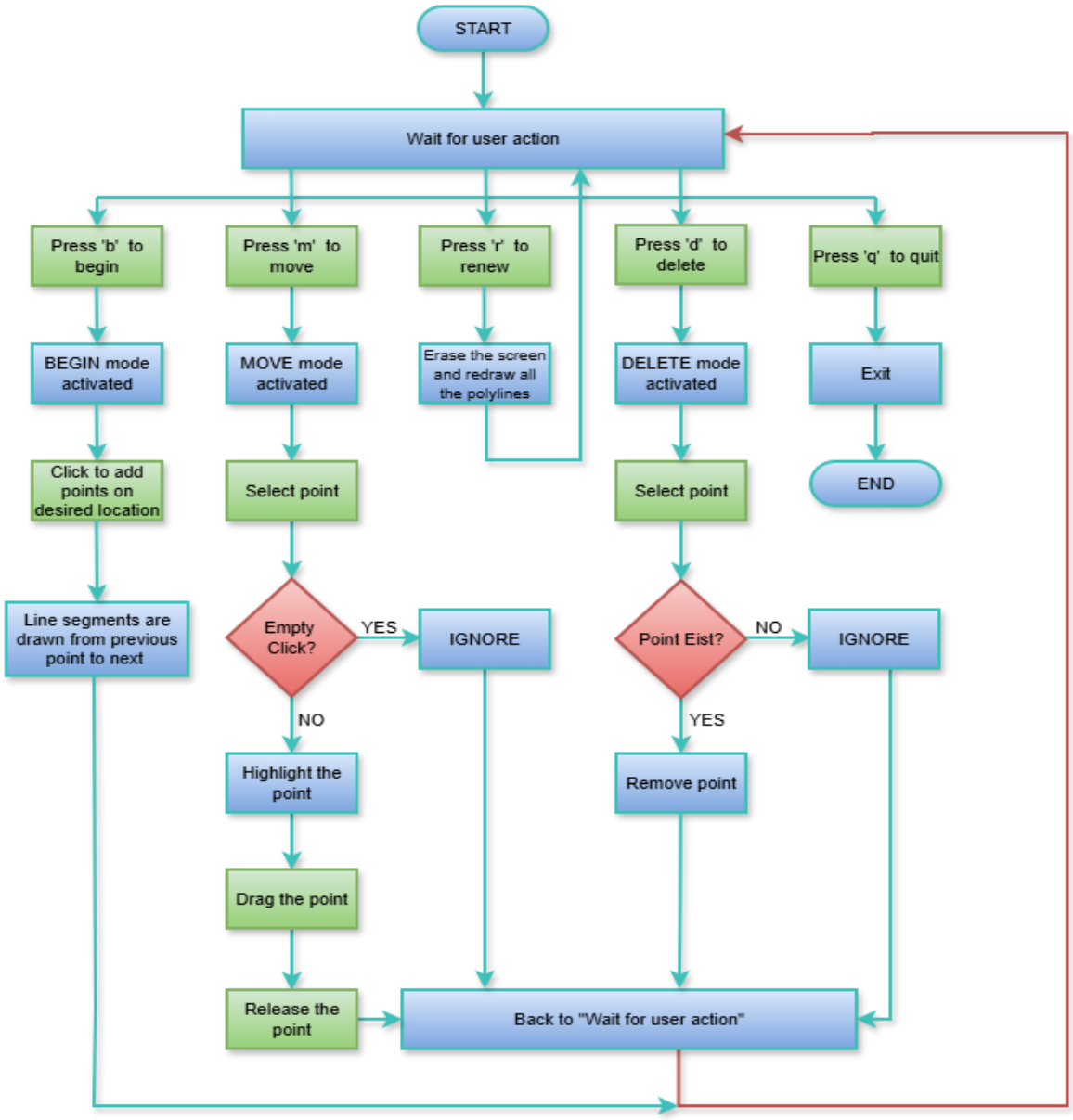
HCI-Based Interaction Design Diagrams

"This section presents visual models of the Polyline Editor system, including flowcharts, use case diagrams, and sequence diagrams. These representations help in understanding user interaction, system behavior, and applied HCI principles."

Tool used : draw.io

1. Flow Chart:

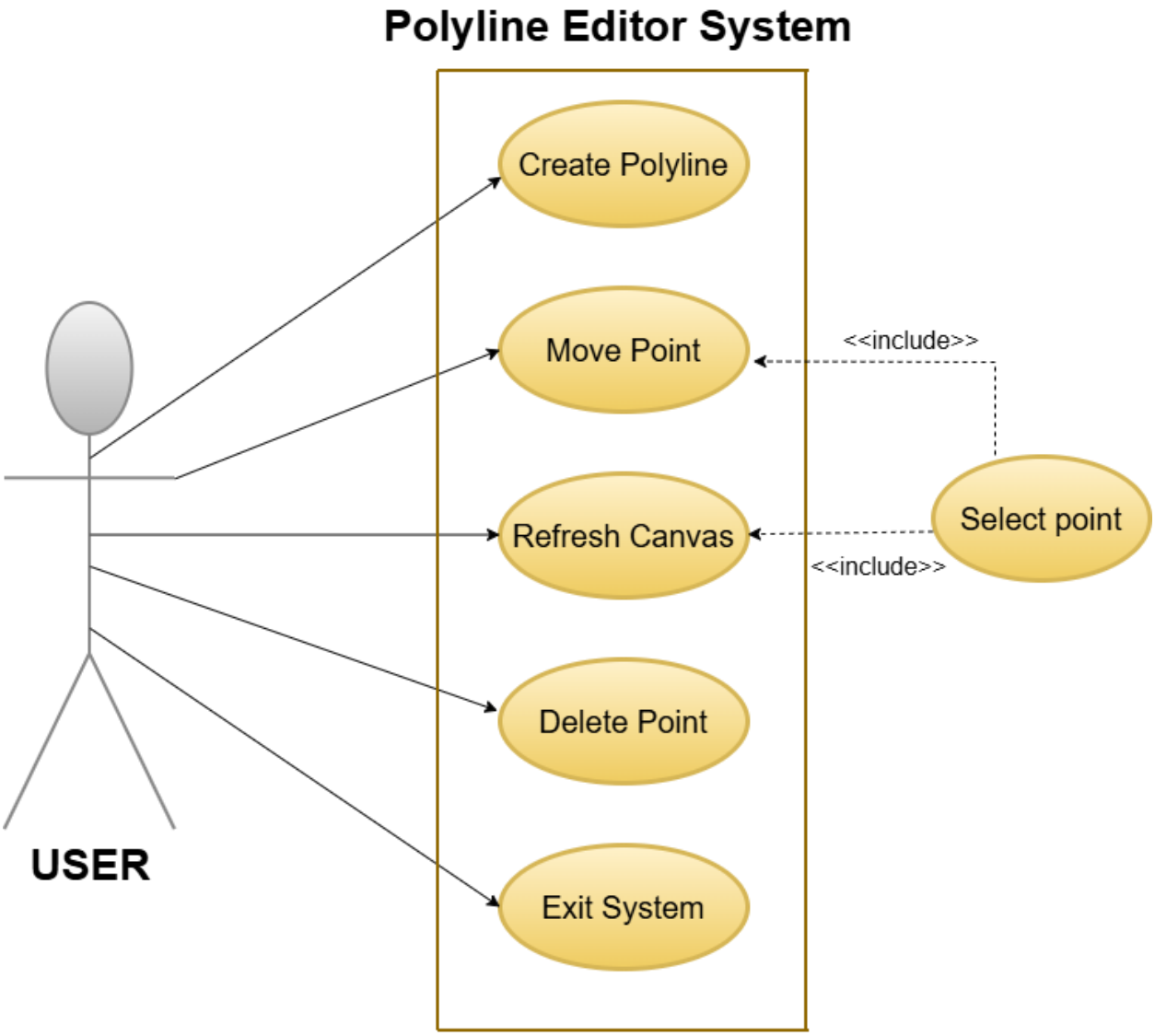


➢ HCI Principles Applied:

- Ensures clear system flow and navigation (user always returns to "Wait for action")
- Supports error prevention (invalid clicks are ignored)

- Provides consistent interaction structure (same pattern for all modes)
- Improves learnability (easy to understand step-by-step process)

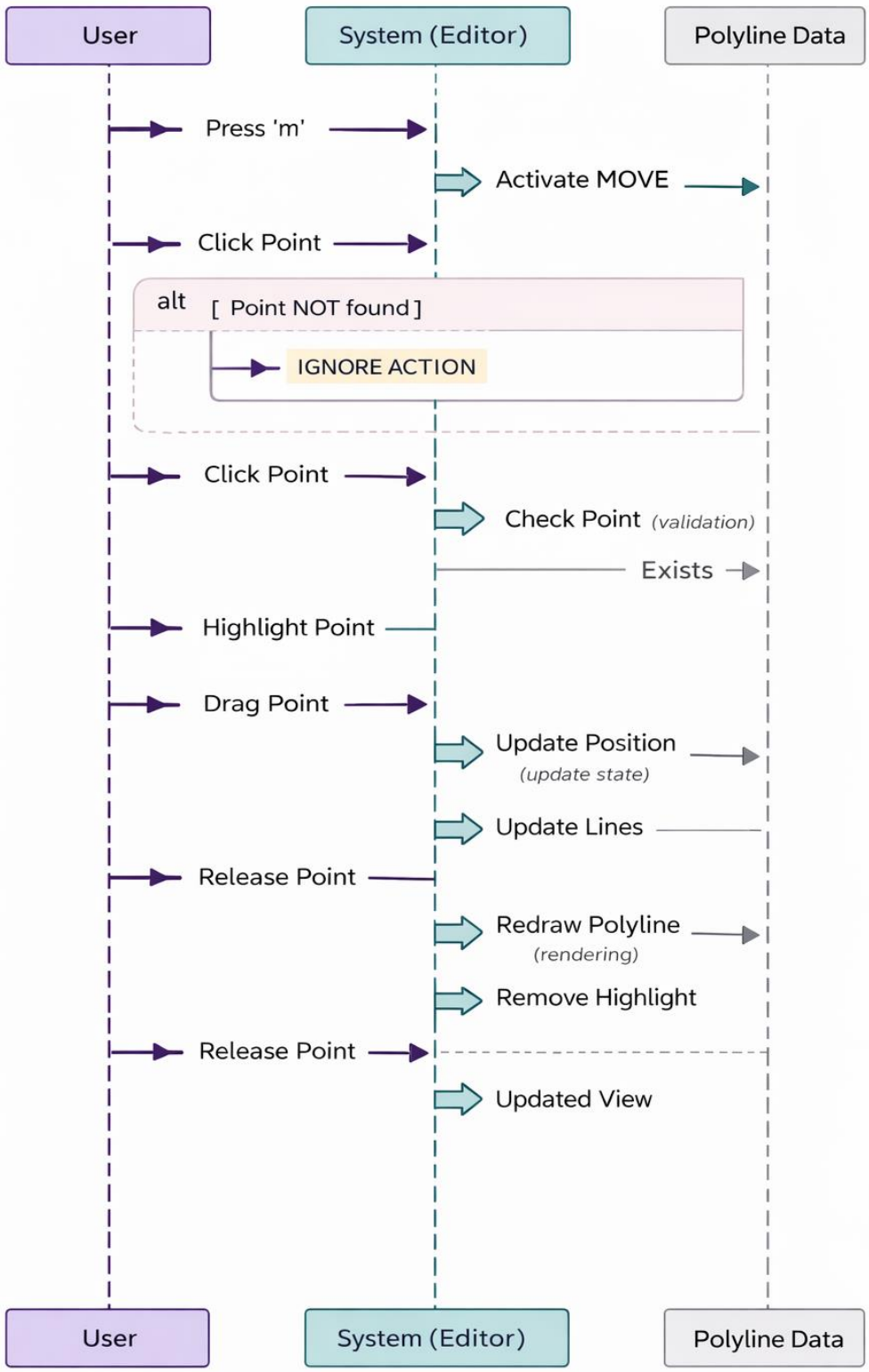
2. Use Case Diagram:



➢ **HCI Principles Applied:**

- Supports user-centered design (focus on user goals like Move, Delete)
- Provides clear task abstraction (actions instead of technical keys)
- Improves simplicity and clarity (easy to understand system functions)
- Ensures consistency (similar structure for all use cases)

3. Sequence Diagram: (Move a Point):



➢ **HCI Principles Applied:**

- Provides immediate feedback (highlight, redraw, update)
- Supports direct manipulation (dragging points visually)
- Ensures visibility of system status (user sees changes instantly)
- Handles error situations gracefully (ignore invalid selection)

4. Analysis Table / Feature Matrix:

"FUNCTIONAL ANALYSIS OF POLYLINE EDITOR INTERACTIONS"

- Shows mapping between user actions and system responses
- Helps identify feedback and error handling

Mode	User Action	System Process	Feedback to User
Begin	Click to add point	Draw line from previous point	Line appears instantly
Move	Select point	Highlight point	Point highlighted
Move	Drag point	Update position & lines	Shape changes live
Move	Release point	Redraw polyline	Final position shown
Delete	Click point	Remove point	Point disappears

Refresh	Press 'r'	Clear & redraw screen	Updated clean canvas
Quit	Press 'q'	Exit system	Application closes

❖ **Overall HCI Considerations:-**

- Feedback
- Consistency
- Error prevention
- User control